

tickets quickly. For example, the upgraded AFC systems in London take 78 seconds to process a ticket compared to the previous system which took 129 seconds, reducing the waiting time for customers by 40 per cent. Commuters' smart cards can be automatically recharged, helping them to avoid long queues for recharging. Self-service kiosks allow commuters to print their own tickets or tokens by submitting the fare to the kiosk. All in all, AFC devices reduce waiting time and effort, multiplying customer satisfaction.

Lower maintenance cost.

Conventional magnetic tokens and paper-based machines had high maintenance costs due to frequent malfunctioning of the system. Paper-based tokens were often lost or damaged. Additionally, commuter throughput was comparatively much slow. Contactless AFC technology helps keep the maintenance cost low. Smart cards cut the expenses on magnetic token maintenance. Customer throughput is also much higher.

For example, Hong Kong's Octopus smart card project for automated fare collection has improved the customer throughput rate through the terminal gates by up to 20 per cent. The system has been implemented across its railway, bus and ferry systems with more than 6200 equipment pieces and 250 distributed computers in place. With transition from magnetic technology to contactless AFC system, the business registered sales of over five million smartcards in 18 months and over \$2.25 billion electronic cash transactions.

Checks frauds. Public transport providers incur financial losses due to fare or ticket evasions. Automated systems allow access to only those customers who have collected the token by making the payment or have sufficient fund balance in their smart cards. Additionally, analytics systems provided by AFC vendors give service providers a clear



Automated fare collection terminals

picture of the total fare collected each day against the number of customers served.

UK Department of Transport studies suggest that AFC systems can cut down fraud travel rates down to 1.5 per cent of the total travels. The UK saves almost \$53 million (40 million pounds) of revenue through automated fare collection.

Back home in India, Mumbai had installed automated fare counter setups for its 2014 Monorail project. The two-phase project, installed by L&T Electrical and Automation, covered 17 stations of the Monorail line. The setup in each station included minimum one ticket vending machine, two ticket readers, three portable ticket decoders, two office machines and eight automatic gates—all channeled to a central server. Commuters have the option to avail smart cards as well as tokens. The setup has dramatically improved the business.

Supporting the cashless era.

Automated faring counters allow multiple payment methods like cash, credit card or debit card. Smart cards can be recharged using additional payment methods like net banking and UPI-based transactions.

Multi-mode payments based on distance, flat payments and collective payment options also enhance the customer convenience.

Integrated system to pay everywhere. Latest AFC systems are not limited to specific services; these can be used across different facilities, e.g., buses, trains and subways. A large majority of passengers favour this system, as it greatly increases the transaction convenience and access. Integrated ticketing system can increase revenues for transport providers by up to 12 per cent. Sydney, Auckland, Melbourne, London, Kuala Lumpur, Singapore and Paris are among cities that have already installed integrated fare systems for transport facilities including buses, railways, subways and ferries.

Who can use this system?

Automated fare collection system finds the biggest implementation in public transportation. Other than Metro rail authorities of different cities, Indian railways and bus and taxi service providers have also successfully adopted AFC technology.

The Delhi Metro Rail Corporation Limited (DMRC) is one of the first and biggest investors in